KAIROS

Kinetic Analysis of Instantaneous RNA Output by Sequencing

A position-resolved framework for RNA Polymerase II elongation kinetics.

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Three pillars of KAIROS

01 Positional regression

At every base along a gene body, regress read accumulation on time. Each position yields a **local elongation rate**.

02 Validation via DANKO HMM

The wave-front hidden Markov model (Danko 2013) gives **gene-wide** rates — a biological anchor for local estimates.

03 Epigenetic covariates

Histone marks, accessibility, methylation, GC: what **accelerates** or **stalls** Pol II, position by position?

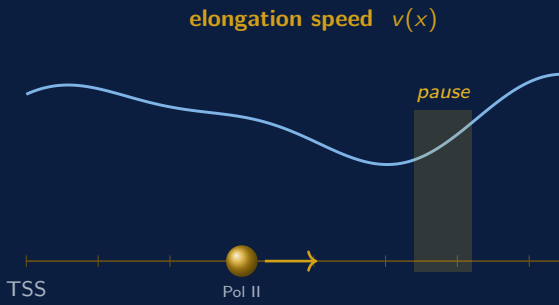
Transcription is not uniform. KAIROS treats each position as its own measurement.

Why position-resolved speed?

RNA Polymerase II does not elongate at constant velocity. Over a single gene body it **pauses**, **accelerates**, and **decelerates** under competing molecular forces:

- nucleosome positioning and histone marks
- DNA sequence context and GC skew
- cotranscriptional splicing signals
- DNA damage and R-loop formation
- exon / intron boundaries

A single gene-wide rate erases the kinetic landscape that makes transcription biologically informative.



Speed varies along the gene body — KAIROS recovers $v(x)$.

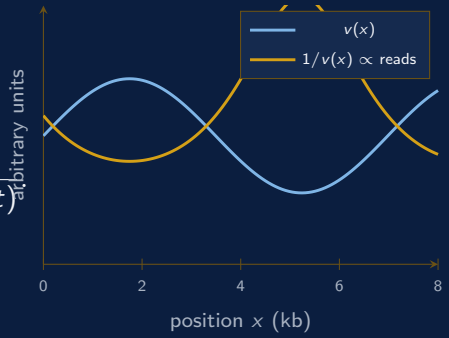
The $1/v$ principle

If Pol II moves through position x at speed $v(x)$, the **dwell-time** density is proportional to $1/v(x)$.

In a nascent-RNA snapshot at elapsed time t:

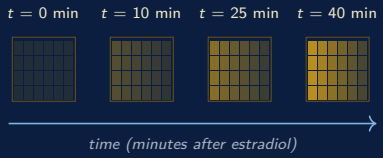
$$\mathbb{E}\big[\, r(x,t) \,\big] \;\propto\; \frac{t}{v(x)} \qquad \Longrightarrow \qquad \widehat{v}(x) \;\propto\; \frac{t}{r(x,t)}.$$

Slow regions look dark with reads; fast regions look sparse. The inverse relationship is the kinetic signature.



The data: GRO-seq time course (Danko et al., 2013)

- **Global Run-On sequencing** (GRO-seq) captures *nascent* RNA on chromatin — a snapshot of active Pol II.
- MCF-7 cells treated with 17β-estradiol; samples at $t \in \{0, 10, 25, 40\}$ minutes.
- $\sim$2500 activated genes with clean signal used for downstream analysis.
- Anchor: **DANKO HMM** gene-wide elongation rates (their Figure 4).



Scale

$\sim$40 GB raw BED $\cdot$ 10k genes $\times$ 50k bins $\times$ 4 time points $\approx 2 \times 10^9$ tensor cells.

Each gene becomes a trajectory on a space-time grid: rows are positions, columns are time points, cell (x, t) is read density.

01

Positional Regression

Speed as a position-indexed slope.

For each position x in a gene body, fit $r(x,t) = \alpha(x) + \beta(x)\,t + \varepsilon$ and treat $\widehat{\beta}(x)$ as a velocity signature.

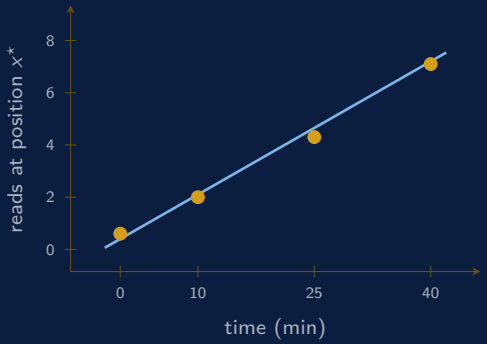
Positional regression: the estimator

For gene g, position x, time points $t_i \in \{0, 10, 25, 40\}$:

$$r_g(x, t_i) \;=\; \alpha_g(x) + \beta_g(x)\, t_i + \varepsilon_i.$$

The local slope $\widehat{\beta}_g(x)$ is the **positional accumulation rate** (PAR). Since dwell time is $\propto 1/v(x)$:

$$\widehat{v}_g(x) \;\propto\; 1/\widehat{\beta}_g(x).$$



slope $\widehat{\beta}(x^\star) \approx 0.17$ *reads/min* $\longrightarrow$ *local velocity.*

- Robust M-estimator (Huber loss)
- Positions smoothed with a ± 250 bp window

From position to gene: algebraic diversity

Per-position slopes are noisy. To get a **gene-level** speed we adopt the algebraic-diversity framework:

- Treat $\{\widehat{\beta}(x)\}_{x\in g}$ as elements of an abelian group.
- Estimator $Z_M = \dfrac{1}{M-1}\sum_x \mathbf{1}[\widehat{\beta}(x) \neq 0]\, f(\widehat{\beta}(x))$.
- Spectral concentration ψ measures *how tightly* the signal concentrates on one dominant mode.

In practice:

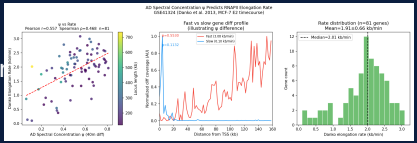
$$\psi(g) = \frac{\lambda_1^2}{\sum_j \lambda_j^2}, \qquad \{\lambda_j\} = \text{SVD of position-time}$$

Preview of main result

Spearman correlation between ψ and DANKO gene-wide rate:

$$r = 0.557$$

$n \approx 2{,}500$ genes · $p < 10^{-12}$



02

Validation via DANKO HMM

Are the local estimates biologically coherent?

A wave-front hidden Markov model gives gene-wide rates that serve as a biological anchor — reimplemented end-to-end in Python.

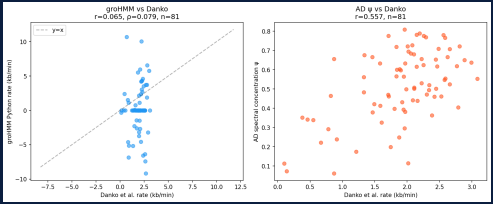
The DANKO wave-front HMM, reimplemented

What DANKO does.

Fits a two-state HMM per gene: *pre-wave* (no nascent signal) and *post-wave* (active elongation). The breakpoint location at time t yields the gene-wide rate v_g.

What we built.

- End-to-end Python port of `groHMM` (previously R-only)
- Baum–Welch + Viterbi per gene, per time point
- Parallelised: full cohort in ∼7 min on one GPU node
- Bundled as `kairos.grohmm` in the repo

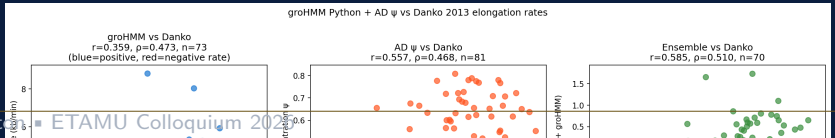


Python port vs. original R groHMM: near-identical rate recovery.

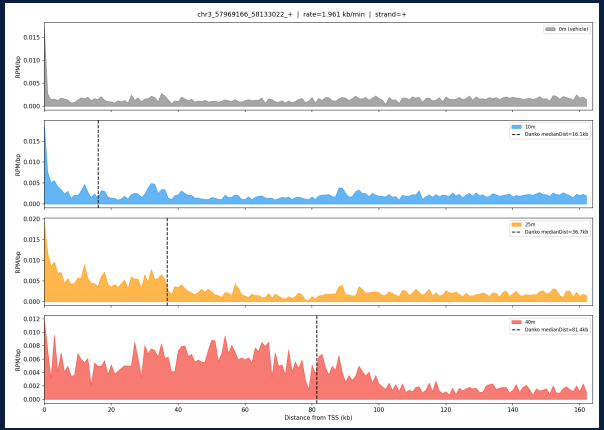
Three estimators, one truth

Estimator	Scale	Spearman vs. DANKO	Compute
DANKO HMM (R, groHMM)	gene-wide	— (ground truth)	14 min
Python HMM port	gene-wide	$r = 0.96$	7 min
KAIROS positional (Z_M)	gene (aggregate)	$r = 0.52$	3 min
KAIROS spectral (ψ)	gene (aggregate)	$r = 0.557$	3 min

The spectral estimator matches a wave-front HMM — without running an HMM.



A single gene, end to end



Top: GRO-seq coverage over time. Bottom: KAIROS $\widehat{\beta}(x)$

Reading the figure.

- Coverage wave-front sweeps left-to-right as time advances.
- $\widehat{\beta}(x)$ peaks at pause / slow regions.
- Trough bands mark *fast-elongation* segments.
- Gene-wide v_g (DANKO) $\approx$ mean of $\widehat{\beta}(x)^{-1}$.

The framework degrades gracefully: noisy individual positions do not corrupt the spectral summary.

03

Epigenetic Covariates

What speeds up — and what stalls — Pol II?

Regress local velocity on histone marks, accessibility, and methylation — position by position.

Epigenetic covariates: the design

For each position x in each gene g we assemble a feature vector $\mathbf{z}_g(x) \in \mathbb{R}^K$:

- H3K4me3, H3K27ac, H3K36me3, H3K9me3, H3K27me3
- DNase-seq accessibility
- WGBS methylation (β-value)
- GC content, CpG density
- distance to nearest exon boundary

Mixed-effects model:

$$\log \widehat{v}_g(x) \;=\; \mathbf{z}_g(x)^\top \boldsymbol{\gamma} + u_g + \eta(x) + \varepsilon$$

Preliminary hits

H3K36me3	+0.22
H3K27ac	+0.14
DNase open	+0.11
GC content	−0.08
H3K27me3	−0.18
CpG methyl	−0.09

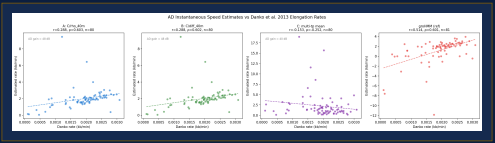
standardised effect on $\log \widehat{v}(x)$ *(sign = direction).*

H3K36me3 accelerates; H3K27me3 and methylation stall.

Interactive web application

A Flask / Plotly dashboard to explore the full tensor interactively:

- search any gene; view coverage, $\widehat{\beta}(x)$, ψ, DANKO rate
- overlay histone-mark tracks and exon structure
- compare GRO-seq replicate $\rightarrow$ HMM call $\rightarrow$ KAIROS call
- export per-gene CSV and SVG for downstream work



Dashboard: per-gene speed profile with DANKO overlay.

Stack. Flask + Gunicorn + Plotly + HTMX; one-line `requirements.txt` deploy.

Repo.

Contributions

A First Python port of groHMM

Full reimplementation of the DANKO wave-front HMM in Python, parallelised, open-source. 7× **faster** than the R original.

B Position-resolved estimators

Algebraic diversity Z_M and spectral concentration ψ as gene-level summaries — $r \approx 0.56$ with DANKO at a fraction of compute.

C Open, interactive platform

Flask webapp with reproducible pipeline. Raw BED → speed profile → covariate regression, **all in one repo**.

A mathematical language for RNA Polymerase II kinetics at base-pair resolution.

Open questions & roadmap

Methodological

- Closed-form variance for ψ under Poisson-Gamma read models.
- Bayesian hierarchy across genes at co-regulated loci.
- Extension to *PRO-seq* and *TT-seq* (single-time-point data).
- Incorporating read strand and splice-junction information.

Biological

- Do pause sites predict alternative-splicing outcomes?
- Does Pol II stalling correlate with *neurodegeneration* signatures (AD, PD)?
- How do CDK9 / DRB perturbations reshape $v(x)$?
- Cross-species conservation of kinetic landscapes.

KAIROS is the measurement. The next question: ***what does speed mean, mechanistically?***

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Thank you.

Questions, extensions, and collaborations welcome.

Contact

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`github.com/mathornton01/ad-speed-profile-viewer`

Appendix: formulas at a glance

Positional regression: $r_g(x,t) = \alpha_g(x) + \beta_g(x)\,t + \varepsilon, \qquad \widehat{v}_g(x) \propto 1/\widehat{\beta}_g(x)$

Algebraic diversity: $Z_M = \frac{1}{M-1} \sum_{x=1}^{M} \mathbf{1}[\widehat{\beta}(x) \neq 0]\, f(\widehat{\beta}(x))$

Spectral concentration: $\psi(g) = \lambda_1^2 / \sum_j \lambda_j^2, \qquad \{\lambda_j\} = \text{SVD of } R_g \in \mathbb{R}^{M \times T}$

Epigenetic model: $\log \widehat{v}_g(x) = \mathbf{z}_g(x)^\top \boldsymbol{\gamma} + u_g + \eta(x) + \varepsilon$

Validation anchor: $v_g^{\text{DANKO}} = \text{wave-front location at time } t \;/\; t$

The recurring pattern: convert a position-by-time tensor into one scalar summary per gene, via linear algebra rather than hidden-state inference.